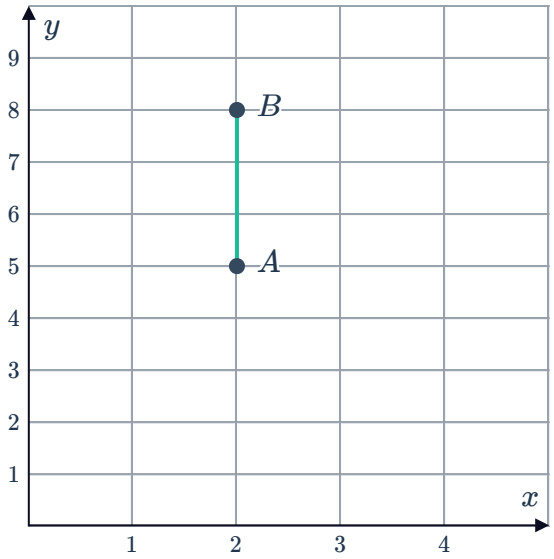
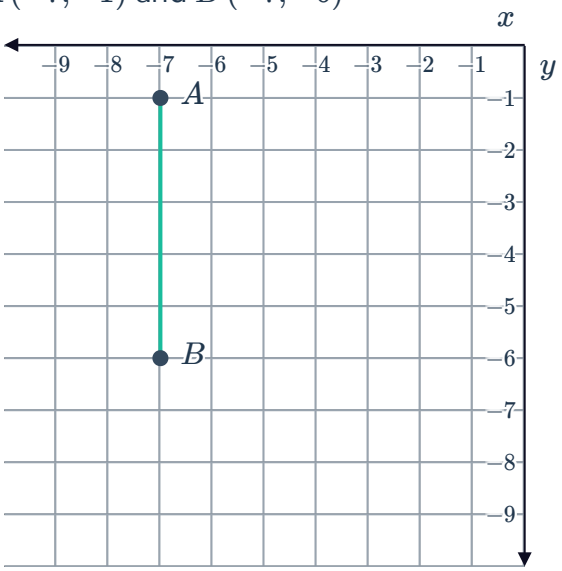
Distance between two points

1 Find the length of each vertical interval on the number planes:

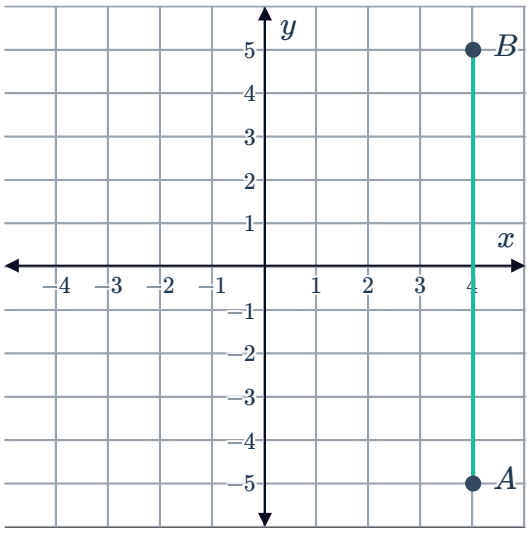
a $A\,(2,5)$ and $B\,(2,8)$



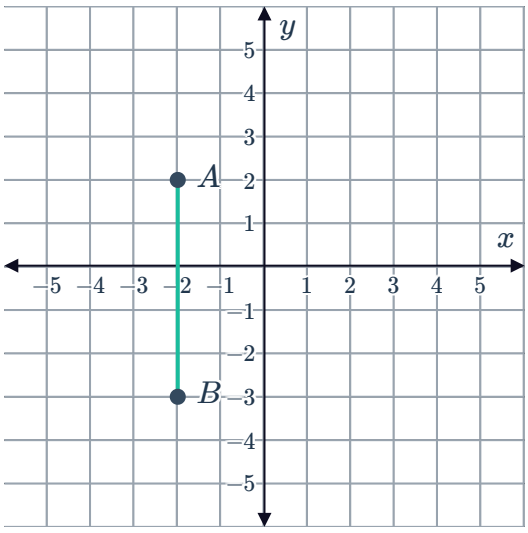
b $A\,(-7,-1)$ and $B\,(-7,-6)$



c $A\,(4,-5)$ and $B\,(4,5)$



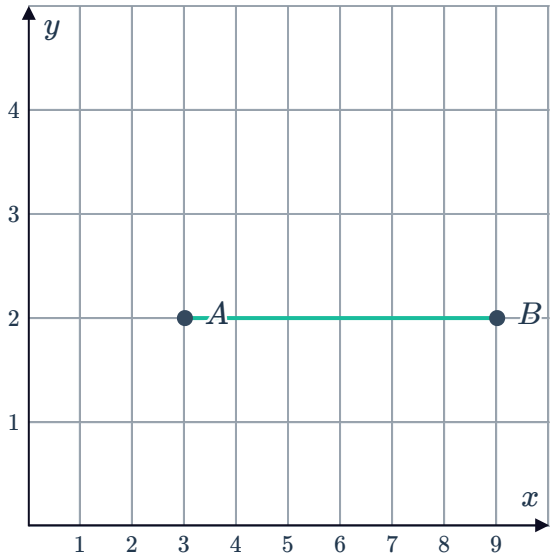
d $A\,(-2,2)$ and $B\,(-2,-3)$



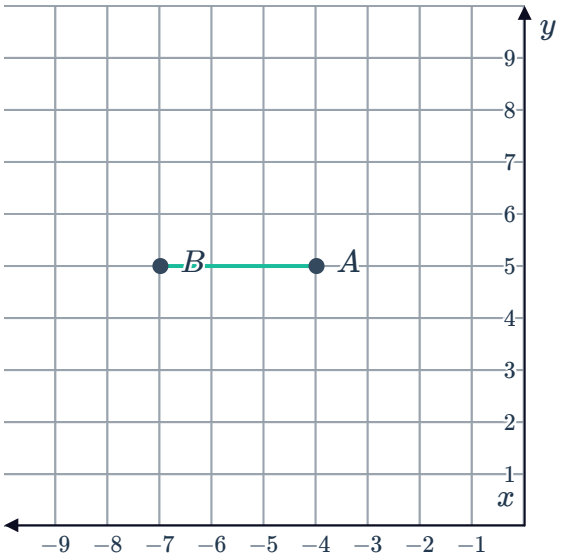
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2 Find the length of each horizontal interval on the number planes:

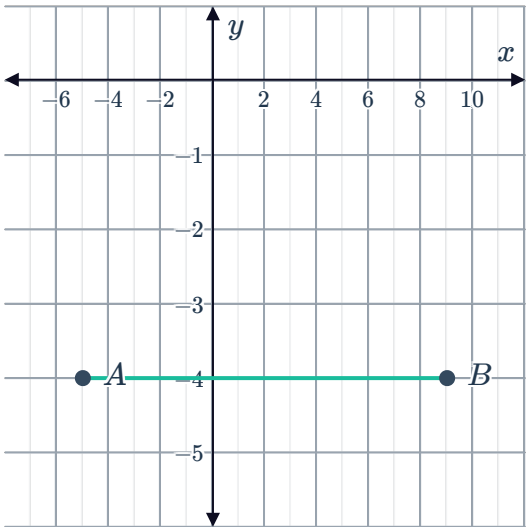
a $A\,(3, 2)$ and $B\,(9, 2)$



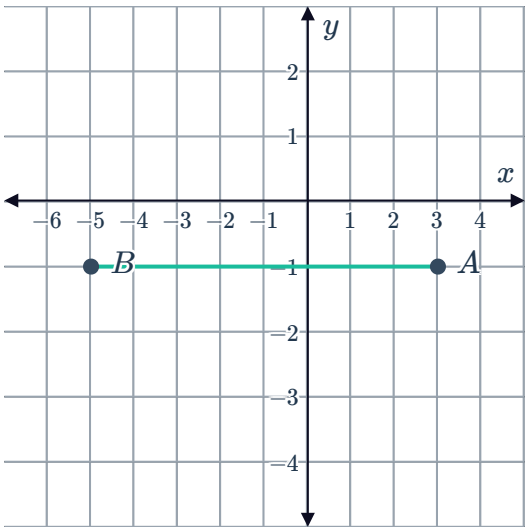
b $A\,(-4, 5)$ and $B\,(-7, 5)$



c $A\,(-5, -4)$ and $B\,(9, -4)$



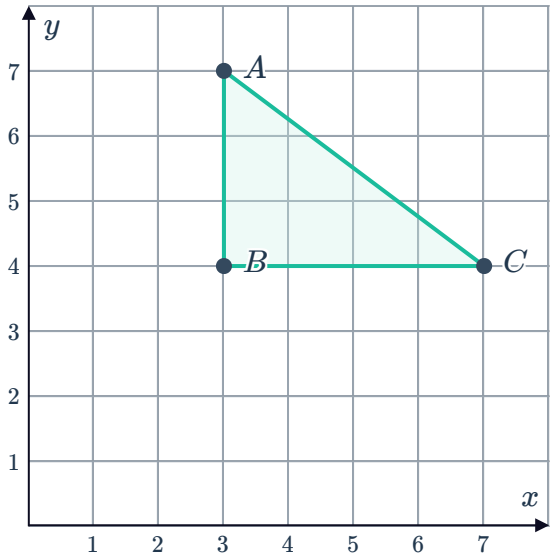
d $A\,(3, -1)$ and $B\,(-5, -1)$



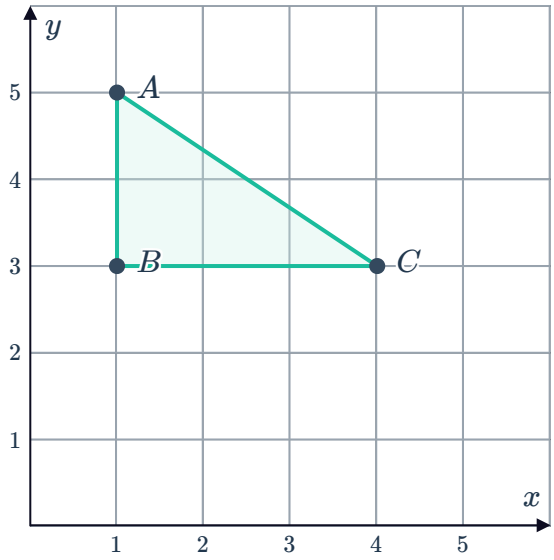
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3 For each of the following right-angled triangles, find the length of side AC. Round your answer to two decimal places if necessary.

a



b



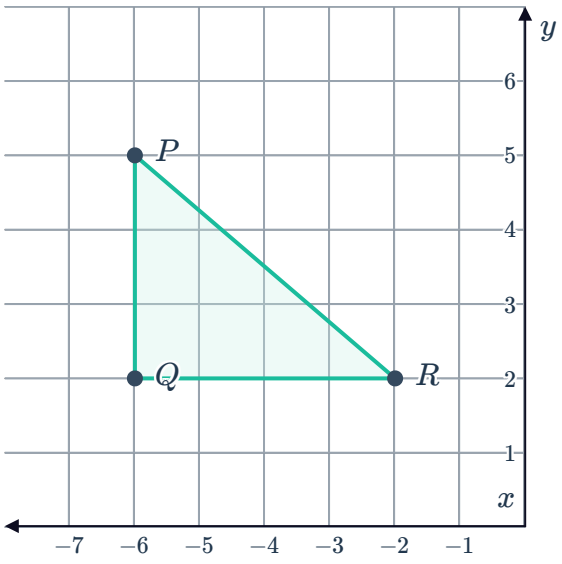
4 For each of the following right-angled triangles:

i Find the length of interval PQ.

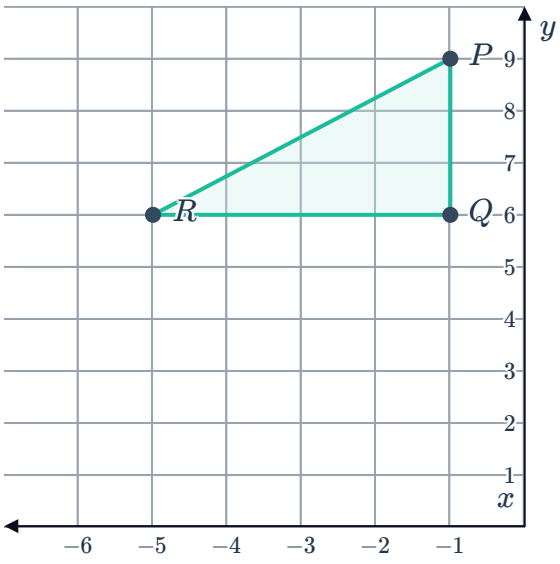
ii Find the length of interval QR.

iii Find the length of PR, rounding to two decimal places if necessary.

a $P\,(-6,5)$, $Q\,(-6,2)$ and $R\,(-2,2)$

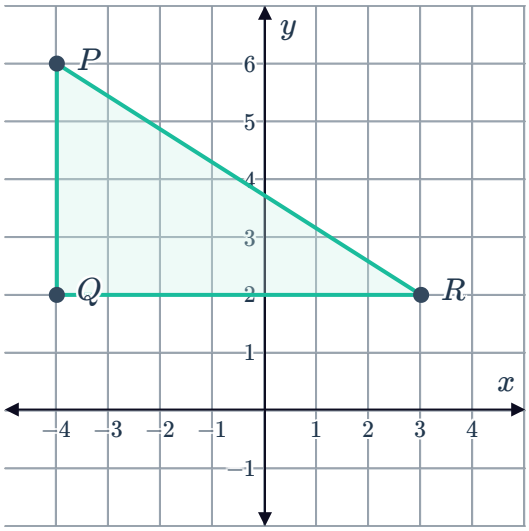


b $P\,(-1,9)$, $Q\,(-1,6)$ and $R\,(-5,6)$

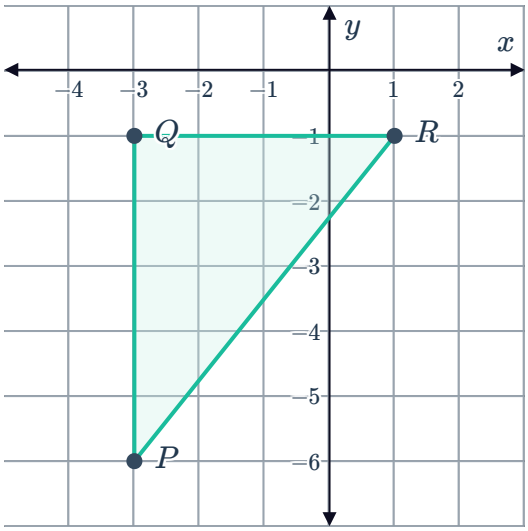


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c $P(-4, 6)$, $Q(-4, 2)$ and $R(3, 2)$



d $P(-3, -6)$, $Q(-3, -1)$ and $R(1, -1)$



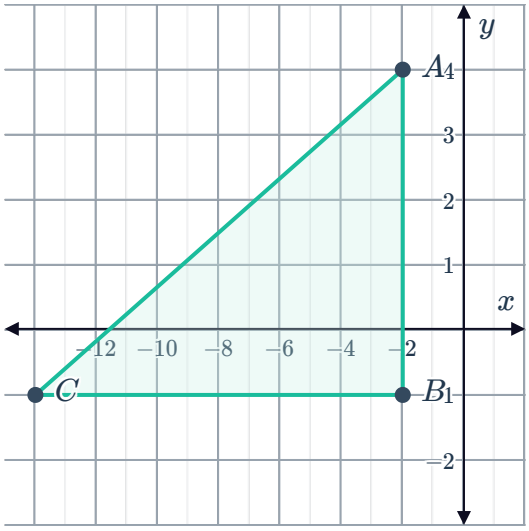
5 For each of the following right-angled triangles:

i Find the length of interval AB.

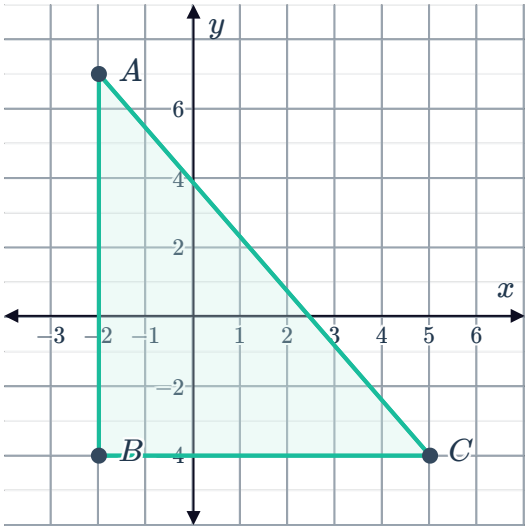
ii Find the length of interval BC.

iii Find the length of AC denoted by c, rounding to two decimal places if necessary.

a $A(-2, 4)$, $B(-2, -1)$ and $C(-14, -1)$

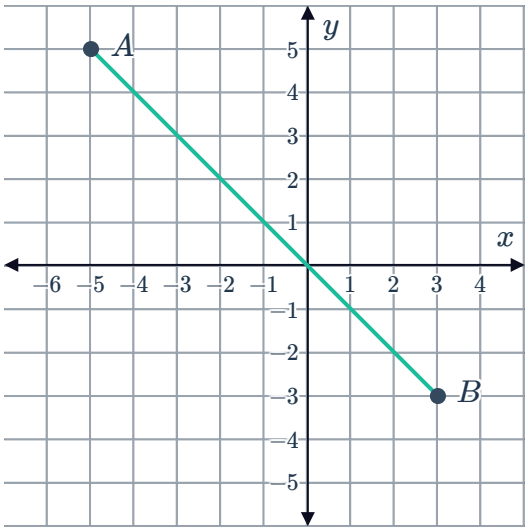


b $A(-2, 7)$, $B(-2, -4)$ and $C(5, -4)$



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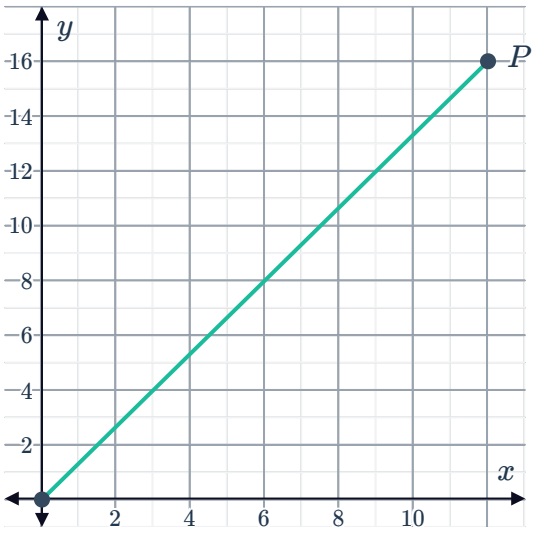
6 Find the length of AB shown on the graph. Round your answer to two decimal places.



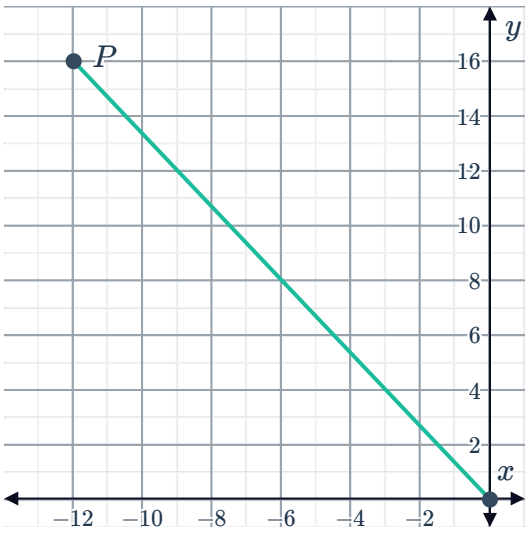
Distance formula

7 Find the distance of the given point P from the origin.

a $P\,(12, 16)$

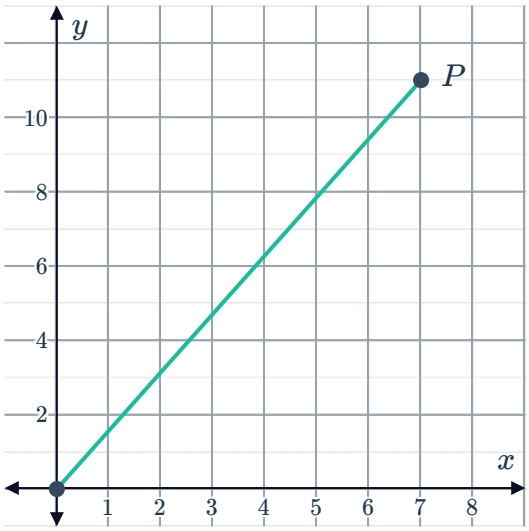


b $P\,(-12, 16)$

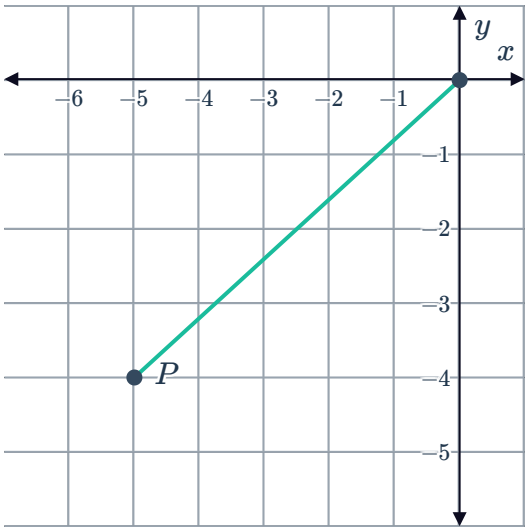


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c $P\,(7, 11)$

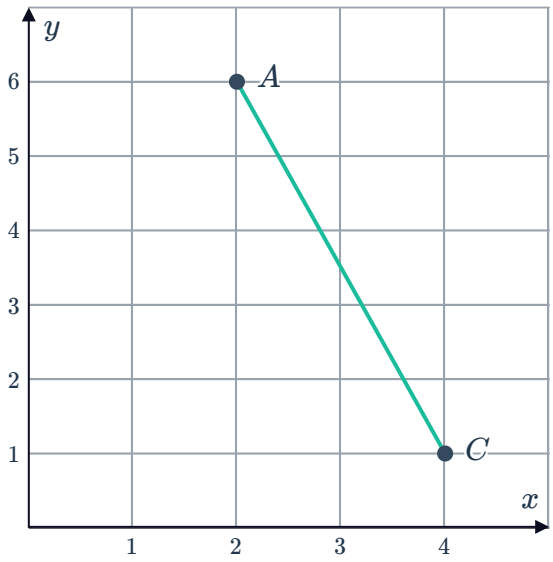


d $P\,(-5, -4)$

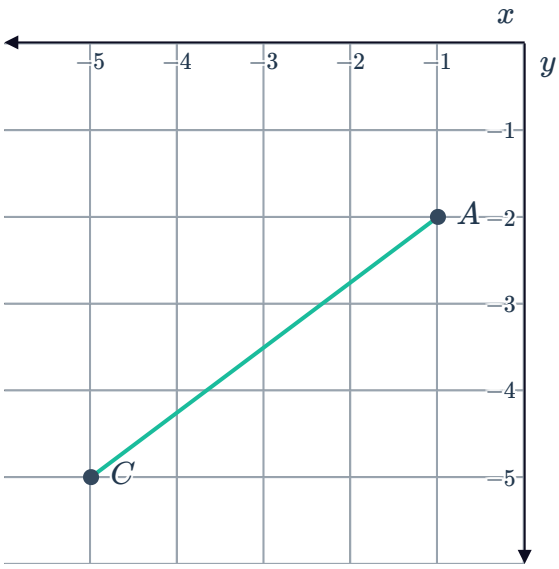


8 For each of the following graphs, find the length of the interval AC. Round your answer to two decimal places if necessary:

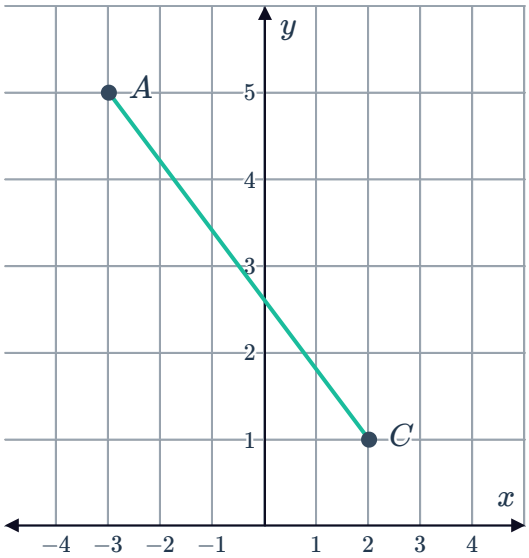
a



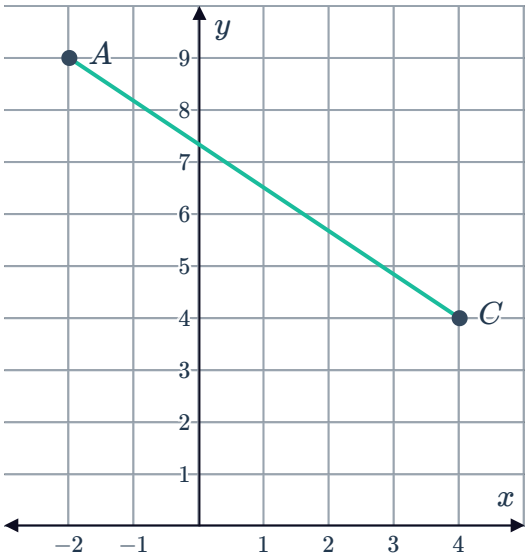
b



c

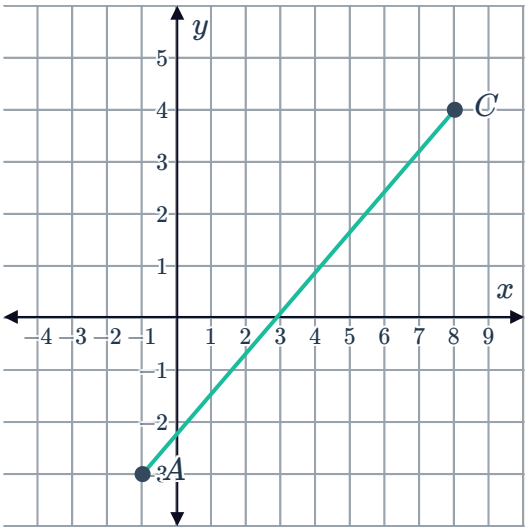


d

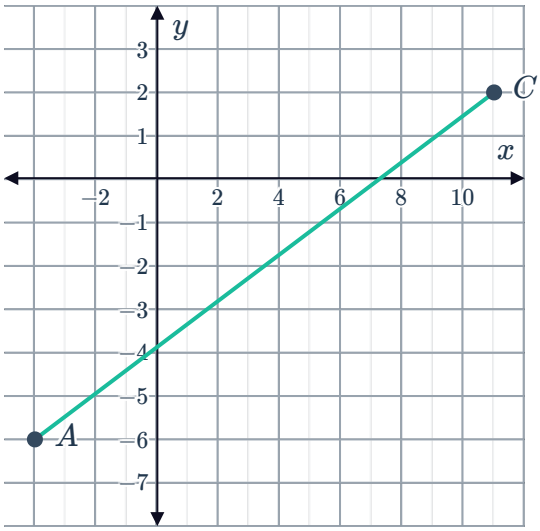


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e



f



9 Find the distance between Point A and Point B. Write your answer in surd form if necessary.

a $A\,(1,4)$ and $B\,(7,12)$

b $A\,(4,2)$ and $B\,(-8,-7)$

c $A\,(-1,9)$ and $B\,(-4,1)$

d $A\left(-1,-\frac{3}{5}\right)$ and $B\left(4,\frac{12}{5}\right)$

10 Consider the points $M\,(-9,-1)$ and $N\,(1,5)$.

a Find the exact distance from M to the origin.

b Find the exact distance from N to the origin.

c Which point is closer to the origin?

11 Given $P\,(4,3)$, $M\,(-3,-4)$ and $N\,(-7,1)$.

a Find the distance from P to M to two decimal places.

b Find the distance from P to N to two decimal places.

c Which point is further from P?

12 Consider the Points $A\,(12,3)$ and $B\,(14,0)$.

a Find the length of AB to two decimal places.

b If M is the midpoint of AB, find the length of AM to two decimal places.

13 AM is a vertical interval 3 units long. If A is the point $(-2,6)$, find two possible coordinates of M.

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14 Consider the points $A\,(-2,-6)$, $B\,(4,-2)$ and $C\,(1,-4)$. Find:

- **a** The exact distance AC.
- **b** The exact distance BC.
- **c** What do you notice about the distance between the points?

15 K is the midpoint of $A\,(3,1)$ and $C\,(15,-7)$. Find the distance from A to K correct to one decimal place.

Geometric applications

16 A triangle is formed by three points: $A\,(4,-3)$, $B\,(1,0)$ and $C\,(7,0)$. Find the following:

- **a** The distance BC.
- **b** The exact distance AB.
- **c** The exact distance AC.
- **d** Is this triangle equilateral, isosceles or scalene?

17 $ABCD$ is a rhombus whose vertices are $A\,(1,2)$, $B\,(3,10)$, $C\,(11,12)$ and $D\,(9,4)$. Find:

- **a** The exact length of diagonals:
  - **i** AC
  - **ii** BD
- **b** The exact area of the rhombus.

18 A circle with centre at point $C\,(3,4)$ has point $A\,(-12,12)$ lying on its circumference. Find:

- **a** The radius of the circle.
- **b** The exact circumference of the circle.
- **c** The distance between point $(20,4)$ and the centre.
- **d** Does the circle also pass through the point $(20,4)$? Explain your answer.

19 A triangle has vertices at $A\,(1,-1)$, $B\,(-3,-4)$ and $C\,(5,-4)$.

- **a** Find the length of the following sides:
  - **i** AB
  - **ii** AC
  - **iii** BC
- **b** Is this triangle equilateral, isosceles or scalene?

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20 Consider the triangle whose vertices are $A\,(9,-2)$, $B\,(2,-8)$ and $C\,(3,5)$.

a How can we show that the triangle has a 90° and two 45° angles?

b Find the length of the following sides in exact form:

i AB **ii** AC **iii** BC

c Does the triangle have a 90° and two 45° angles?

21 $\triangle PQR$ has vertices $P\,(2,-6)$, $Q\,(-9,-17)$ and $R\,(-5,1)$.

a Find the length of the following sides in exact form:

i PQ **ii** QR **iii** PR

b Is the triangle right-angled?

22 The points $M\,(-3,-5)$ and $R\,(4,-12)$ are the end points of an interval. $N\,(-1,-7)$ is a point on this interval.

a Find the exact distance between the following points:

i M and N **ii** N and R

b In what ratio does N divide MR?

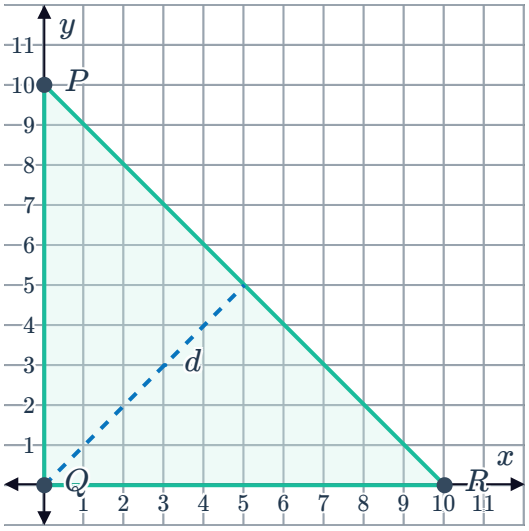
23 Find the perimeter of the parallelogram whose vertices are $A\,(3,2)$, $B\,(-2,7)$, $C\,(2,9)$ and $D\,(7,4)$. Round your answer correct to one decimal place.

24 The isosceles $\triangle PQR$ is shown on the number plane:

a Find the area of the triangle.

b Find the exact length of PR.

c Find the exact value of d, the perpendicular distance from Q to the side PR.



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25 Consider the points $P\,(10x, -6x)$ and $Q\,(4x, 2x)$, where $x > 0$.

- **a** Find the distance between P and Q in terms of x.
- **b** Find the coordinates of the midpoint of the segment PQ in terms of x.

26 $P\,(-4, -4)$, $Q\,(5, -7)$, $R\,(9, -11)$ and $S\,(0, -8)$ are the vertices of a quadrilateral.

- **a** Find the exact length of side PQ.
- **b** Find the exact length of side QR.
- **c** Find the exact length of side RS.
- **d** Find the exact length of side SP.
- **e** Find the exact length of the diagonal PR.
- **f** Find the exact length of the diagonal QS.
- **g** Classify the quadrilateral.

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